

E-Commerce Management System

Module: Shipping Management Module — End User & Technical Documentation

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GitHub Repo: ansarizeenat/ecommerce-category-management	Live Server (Render): https://ecommerce-category-management.onrender.com
Tech Stack: Java 25, Spring Boot 4.1, JPA, MySQL 8.0, Thymeleaf	Date: September 2026

1. Introduction

The **Shipping Management Module** handles the end-to-end shipping and fulfillment process, including calculating dynamic shipping costs, tracking shipments with real-time milestones, and managing courier logistics providers. It guarantees that customer orders are dispatched, tracked, and delivered efficiently, accurately, and on schedule.

2. Database Design & Specification

Table Name: **shipping** — schema exactly matches the Itvedant assignment specification.

Table 1: shipping

Column Name	Data Type	Key / Constraint	Description
shipping_id	INT	PK, Auto-Increment	Unique identifier for each shipping entry
order_id	INT	FK (orders.id)	Foreign key referencing the orders table
courier_service	VARCHAR(100)	NOT NULL	Name of courier or shipping provider (e.g. Blue Dart, FedEx, DHL)
tracking_number	VARCHAR(100)	NOT NULL, UNIQUE	Unique tracking number provided by the courier
shipping_status	VARCHAR(50)	NOT NULL	Status of the shipment (Shipped, In Transit, Delivered)
shipping_cost	DECIMAL(10,2)	NOT NULL	Shipping cost calculated for the order
created_at	DATETIME	AUTO-GENERATED	Timestamp when shipping was initiated
updated_at	DATETIME	AUTO-UPDATED	Timestamp when shipping details were last updated

3. Key Functionalities

3.1 Shipping Cost Calculation

Shipping costs are dynamically calculated based on weight, delivery location, and shipping method:

- **Weight Fee:** Base fee of \$5.00 plus \$2.00 per kg.
- **Delivery Location:** Local (1.0x, 1-2 days), Domestic (1.4x, 3-5 days), International (2.5x, 7-10 days).
- **Shipping Method:** Standard (1.0x), Express (1.5x), Overnight (2.2x).
- **Formula:** Total Cost = (Base Cost + Weight x \$2.00) x Location Multiplier x Method Multiplier.

An interactive cost calculator is provided in the admin UI and exposed via POST /api/shipping/calculate-cost.

3.2 Shipping Dashboard

Admins can monitor all orders with corresponding shipping details: Shipping ID, Order ID, Customer Name, Address, Courier Service, Tracking Number, Status Badge, Cost, and Dispatch Date. Includes real-time metric counters for Total Shipments, Shipped, In Transit, and Delivered, along with interactive status filter buttons.

3.3 Track Shipment (Customer Self-Service)

Customers and admins can look up any shipment status by entering its tracking number into the tracking search box or clicking the tracking number link on any table. The system renders an interactive milestone progress stepper showing: 1. Order Confirmed • 2. Shipped • 3. In Transit • 4. Delivered, with timestamps and destination address.

3.4 Update Shipping Information

Admins can update shipping details once an order has been shipped, including re-assigning courier service (e.g. to Blue Dart or FedEx), updating tracking numbers, and changing shipment status (Shipped → In Transit → Delivered). Updating status to 'Delivered' automatically synchronizes the parent order's status to Delivered.

4. How to Use the Screen

Admin Flow:

1. Open the application and click the **Shipping Management** tab.
2. In the *Calculate Cost & Dispatch Order* form, select an active order.
3. Choose courier service (Blue Dart, FedEx, DHL, Delhivery, Speed Post) and adjust weight, location, and method.
4. Click **Create Shipment & Dispatch**. The order transitions to 'Shipped' and appears in the dashboard.
5. Click the **Pencil** icon on any row to edit courier details, tracking number, or advance the status to Delivered.

Customer Tracking Flow:

1. Enter the tracking number (e.g. BD-83920145) into the *Track Shipment* search box.
2. Click **Track** to view live shipment progress, destination address, and milestone timeline.

5. REST API Specifications

HTTP Method	Endpoint	Description
GET	/api/shipping	List all shipments (supports ?status=Shipped/In Transit/Delivered)
GET	/api/shipping/stats	Summary counters (Total, Shipped, In Transit, Delivered, Total Cost)
GET	/api/shipping/{id}	Retrieve single shipping record by ID
GET	/api/shipping/order/{orderId}	Retrieve shipping details for an order
GET	/api/shipping/track/{trackingNumber}	Public shipment tracking endpoint with milestone timeline
POST	/api/shipping/calculate-cost	Calculate shipping cost based on weight, location, and method
POST	/api/shipping	Create shipment and update linked order status to Shipped
PUT	/api/shipping/{id}	Update courier service, tracking number, or shipping status
PUT	/api/shipping/{id}/status	Quick status update (Shipped, In Transit, Delivered)

6. Assignment Submission Deliverables

Assignment Requirement	Candidate Submission Details
1. GitHub Repository Link	https://github.com/ansarizeenat/ecommerce-category-management
2. URL Hosted on Free Server	https://ecommerce-category-management.onrender.com
3. End User Documentation	Compiled in PDF: Shipping_Management_Module_End_User_Documentation.pdf